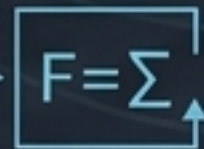


Gas-and-Water-Coning

Mechanisms,



Critical Production Rates,
and Empirical Correlations

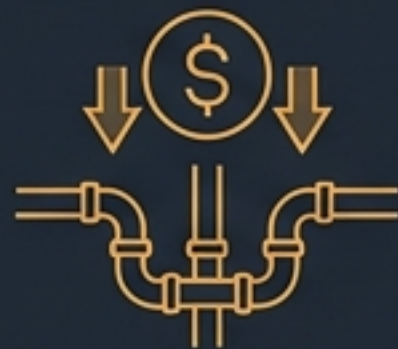


Prepared by
Engineer Sajjad K. Hano

The Impact of Coning on Reservoir Recovery



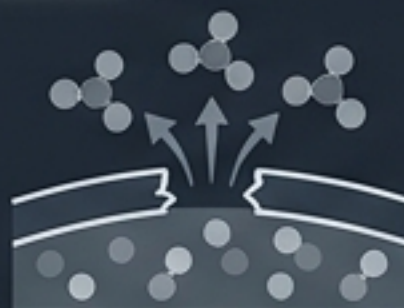
ECONOMIC DRAIN



Costly added water and gas handling infrastructure.



PRESSURE DEPLETION

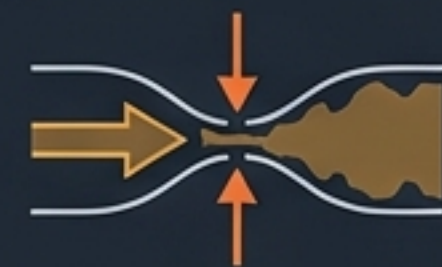


GAS ESCAPE

Gas production from the cap rapidly reduces primary gas drive pressure.



EFFICIENCY LOSS



REDUCED FLOW

Severe **reduction** in the **efficiency** of the reservoir's depletion mechanism.



CORROSION DAMAGE



Affliction by **corrosive water** necessitates **expensive disposal and treatment**.

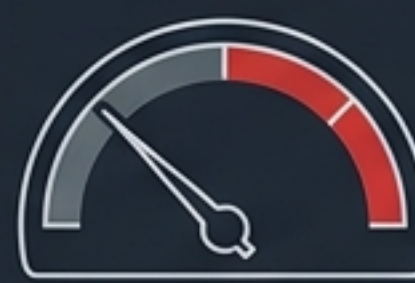


ABANDONED WELL



PREMATURE DEATH

Early, **forced abandonment** of afflicted producing wells.



ULTIMATE LOSS

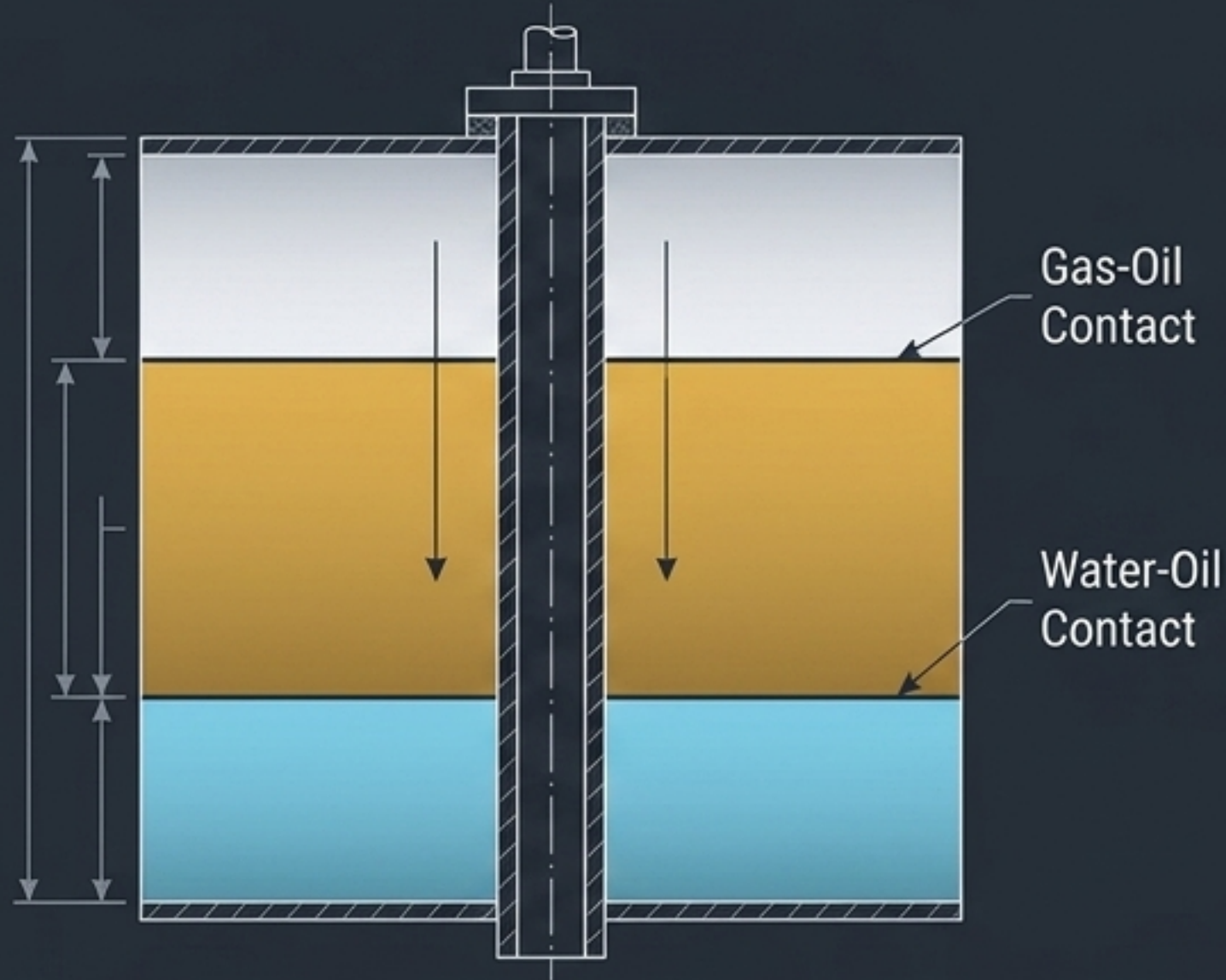


LOST OIL

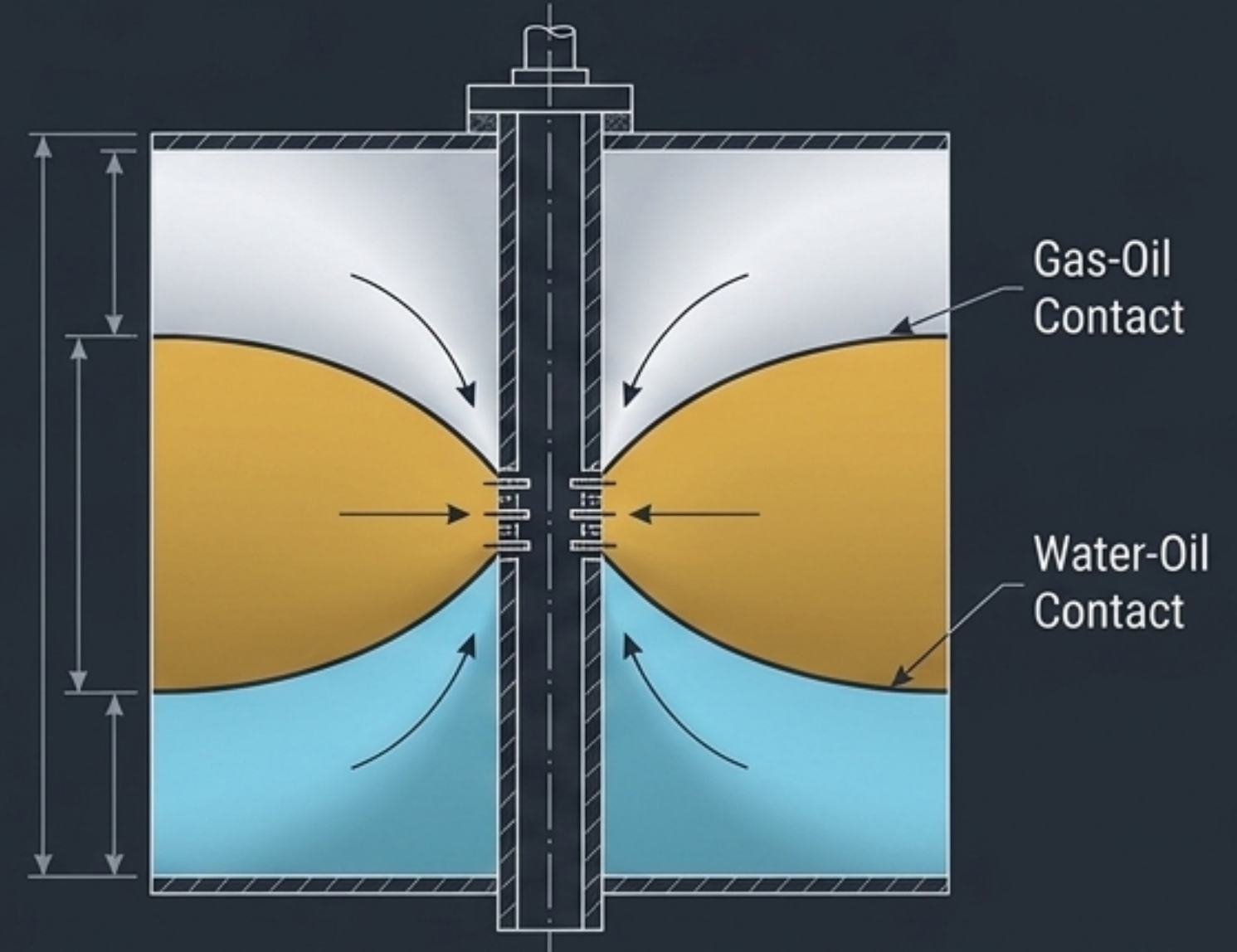
Significant reduction in total field overall **oil recovery**.

Static Equilibrium vs. Dynamic Deformation

Static Condition (Gravity Equilibrium)

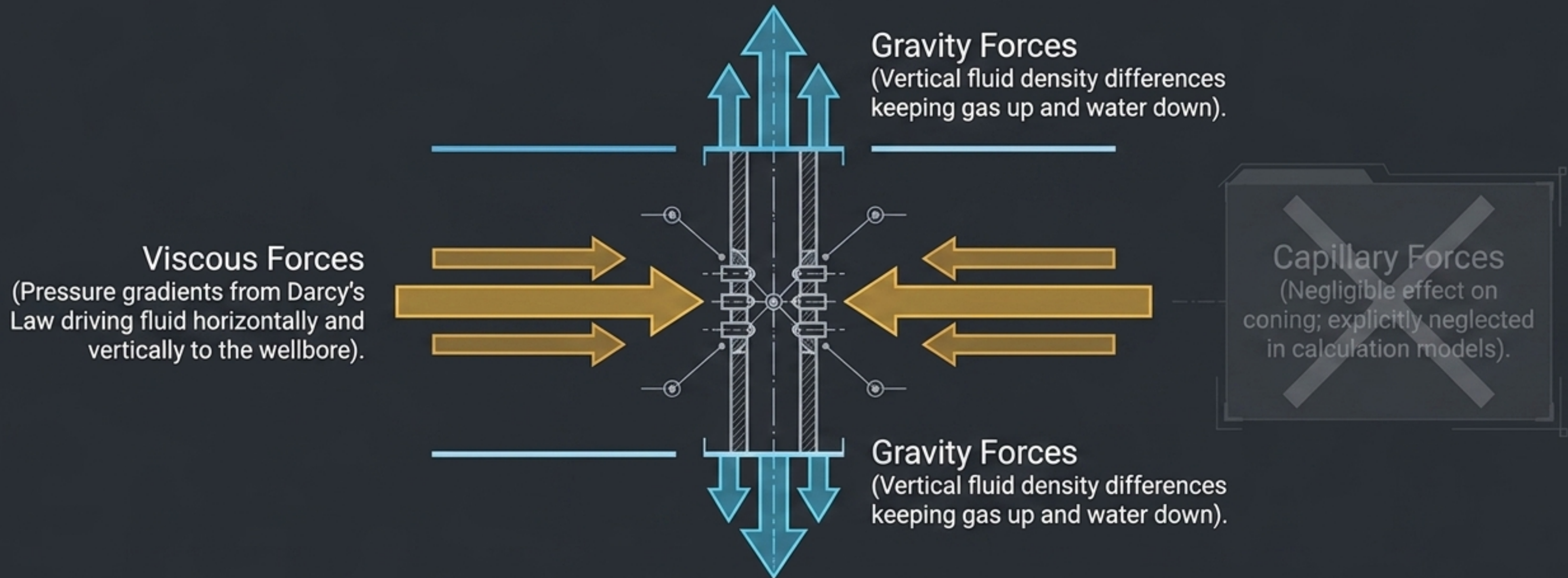


Dynamic Coning (Flowing Gradients)



Key Insight: Coning is the upward movement of water and/or downward movement of gas into producing perforations. Reservoir fluids move in the direction of least resistance, disrupting original gravity equilibrium.

The Mechanics of Coning: A Battle of Forces



At any given time, fluid flow distribution is dictated by the balance between dynamic viscous forces and static gravitational forces. When viscous pressure drops at the wellbore exceed gravitational separation forces, a cone breaks into the well.

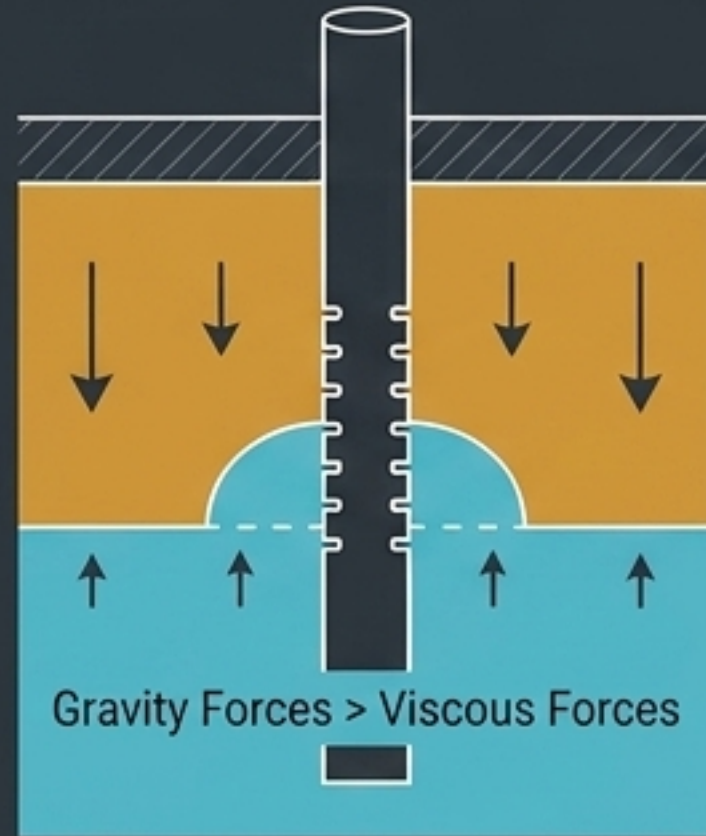
Defining Cone Stability Thresholds

Stable Cone

Condition: Viscous Forces < Gravity Forces

Behavior: A cone forms but does not extend to the well. It neither advances nor recedes.

State: Steady-state condition is achieved.

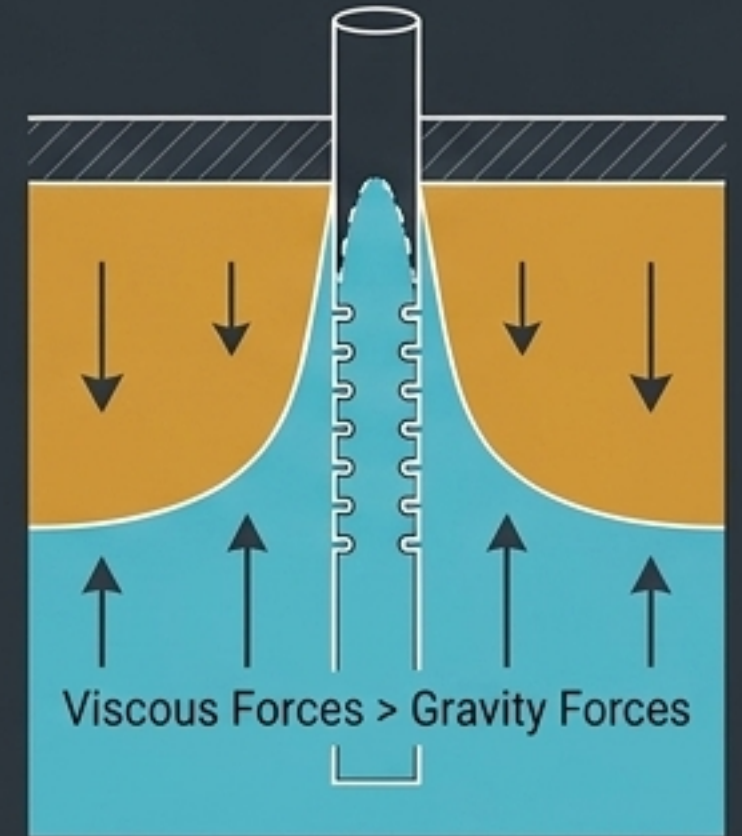


Unstable Cone

Condition: Viscous Forces > Gravity Forces

Behavior: The flowing pressure drop overcomes gravity. The cone grows and ultimately breaks into the wellbore.

State: Unsteady-state condition (until equilibrium is reached post-breakthrough).

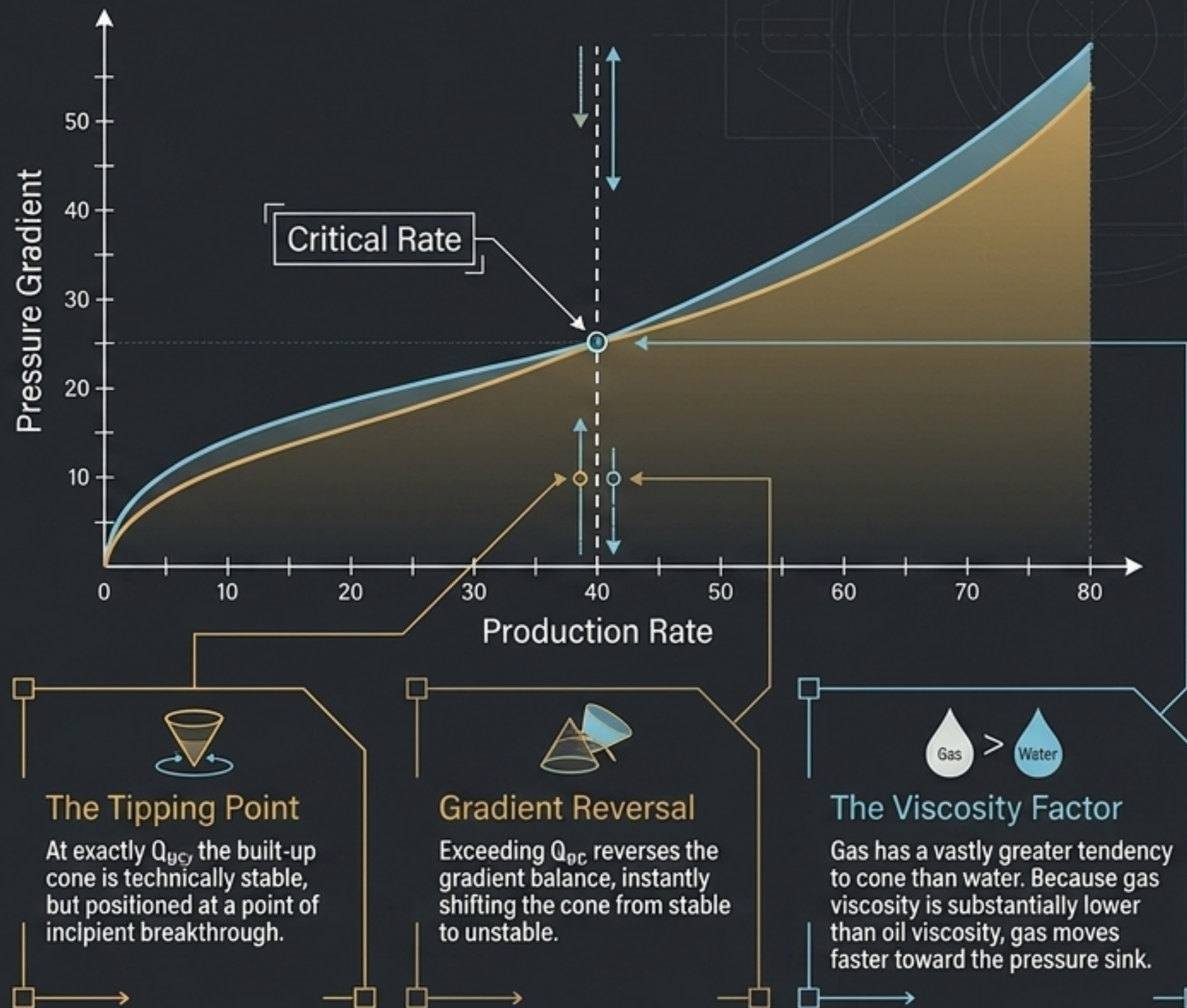


The "Pseudo-Stable" Reality: A stable system is rarely permanent. Reservoir depletion causes the water-oil contact to advance, and well damage increases the required pressure drop. Both inevitably push a stable cone into an unstable state.

The Critical Production Rate (Q_{oc})



The absolute maximum rate of oil production achievable without concurrent production of the displacing phase (water or gas).



Engineering Solutions: Three Categories of Correlation

The Meyer-Garder Correlation (1954)

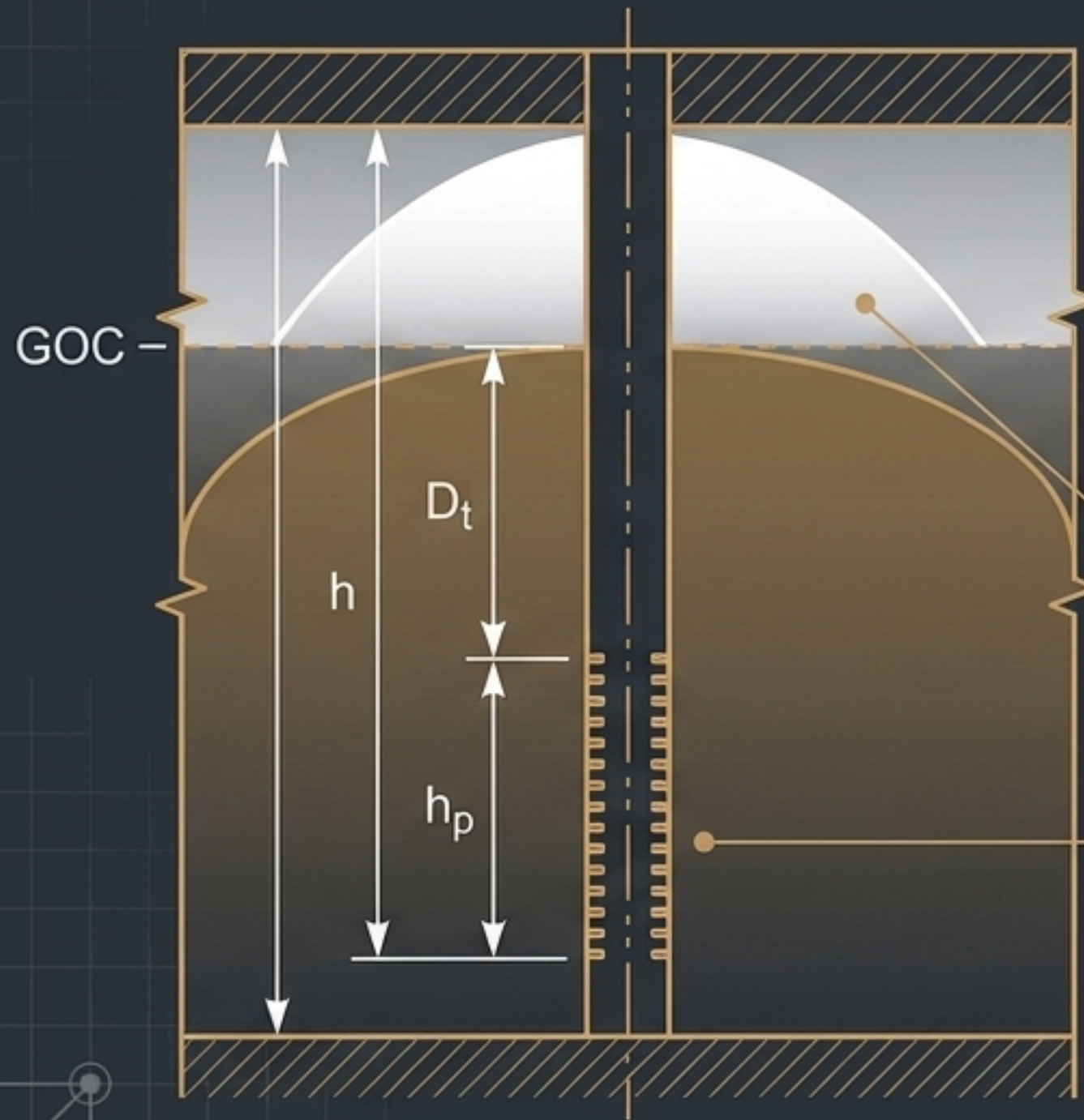
- **Assumption:** A homogeneous reservoir system with uniform horizontal and vertical permeability ($k_h = k_v$).
- **Driver:** Driven by the radial flow of oil and the associated pressure sink around the wellbore.

Critical Rate Calculations
(Our Diagnostic Focus)

Breakthrough Time Predictions

Well Performance Calculations Post-Breakthrough

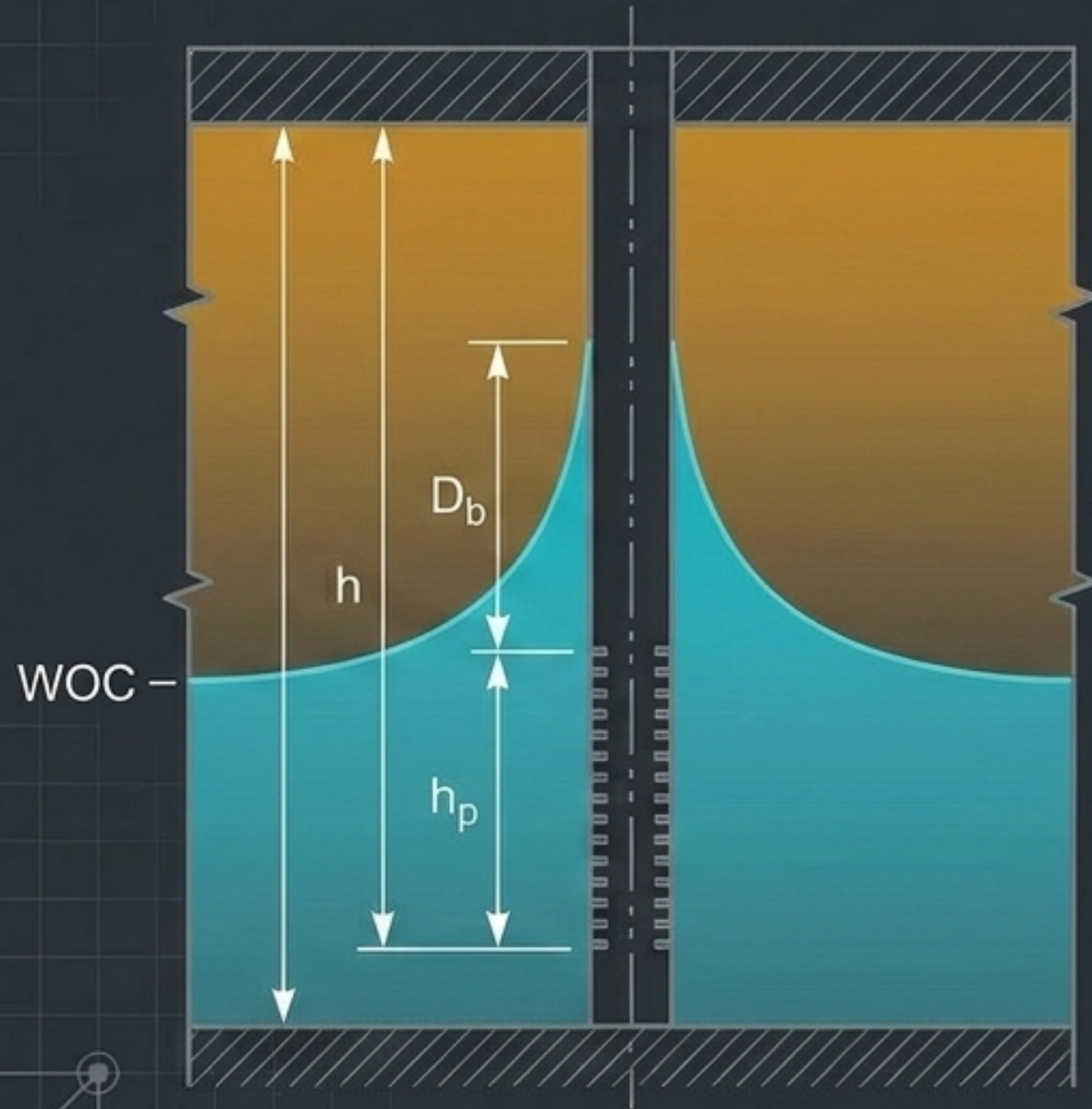
Meyer-Garder Critical Rate: Gas Coning



$$Q_{oc} = 0.246 \cdot 10^{-4} \left[\frac{p_o - p_g}{\ln\left(\frac{r_e}{r_w}\right)} \right] \left(\frac{k_o}{u_o B_o} \right) \left[h^2 - (h - D_t)^2 \right]$$

- **Gravity Driver ($p_o - p_g$):** The density difference between oil and gas directly dictates the gravitational pushback against coning.
- **Geometric Limits:** The safe production rate is exponentially constrained by the depth to the top of the perforations (D_t) relative to the total oil column (h).

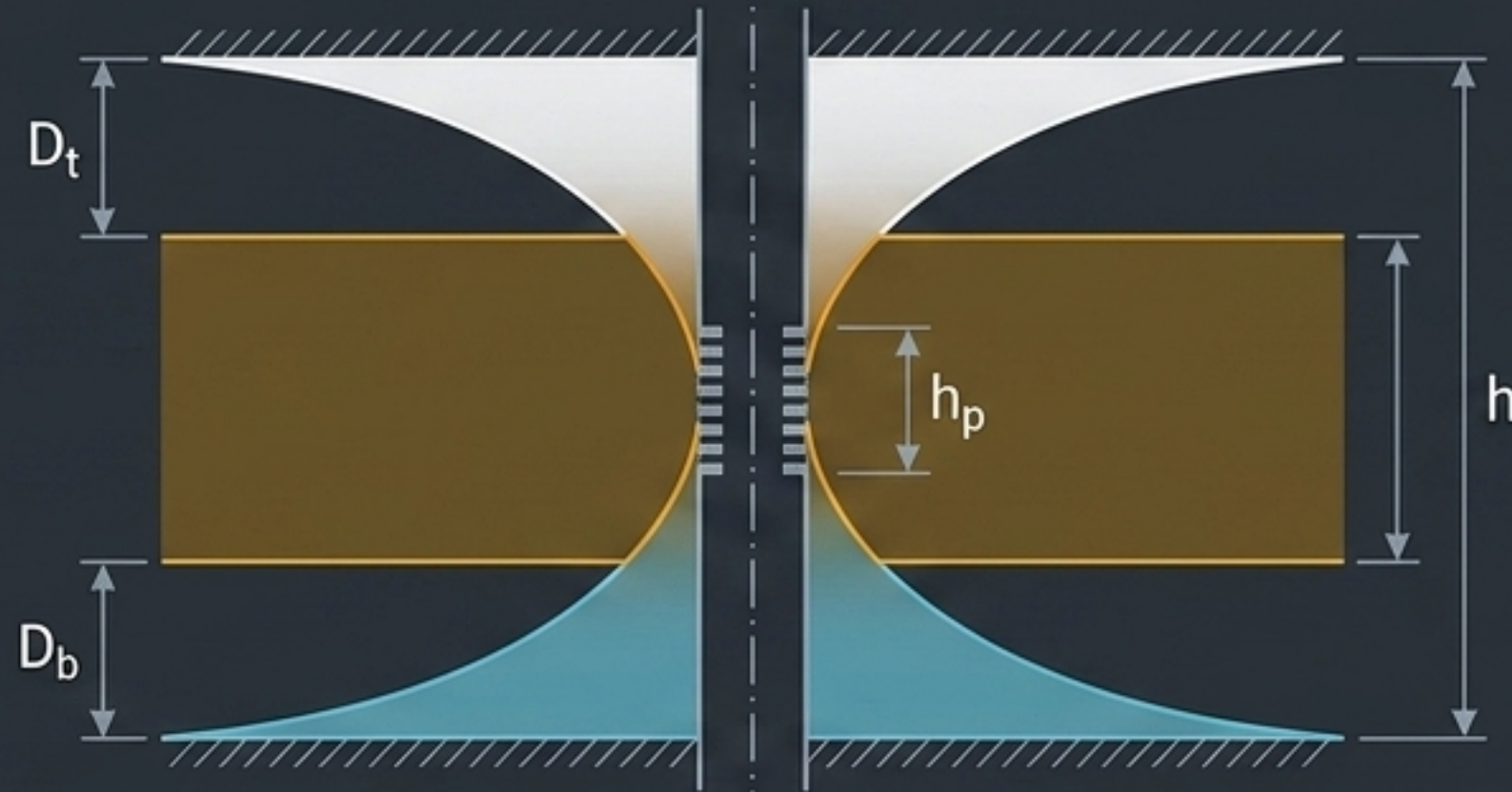
Meyer-Garder Critical Rate: Water Coning



$$Q_{oc} = 0.246 \cdot 10^{-4} \left[\frac{p_w - p_o}{\ln\left(\frac{r_e}{r_w}\right)} \right] \left(\frac{k_o}{u_o B_o} \right) (h^2 - h_p^2)$$

Structural Observation: The architecture of this correlation mirrors the gas system perfectly. The mathematical adjustments simply swap gas density for water density (p_w), and isolate the perforated interval (h_p) relative to the underlying water-oil contact.

The Ultimate Challenge: Simultaneous Coning

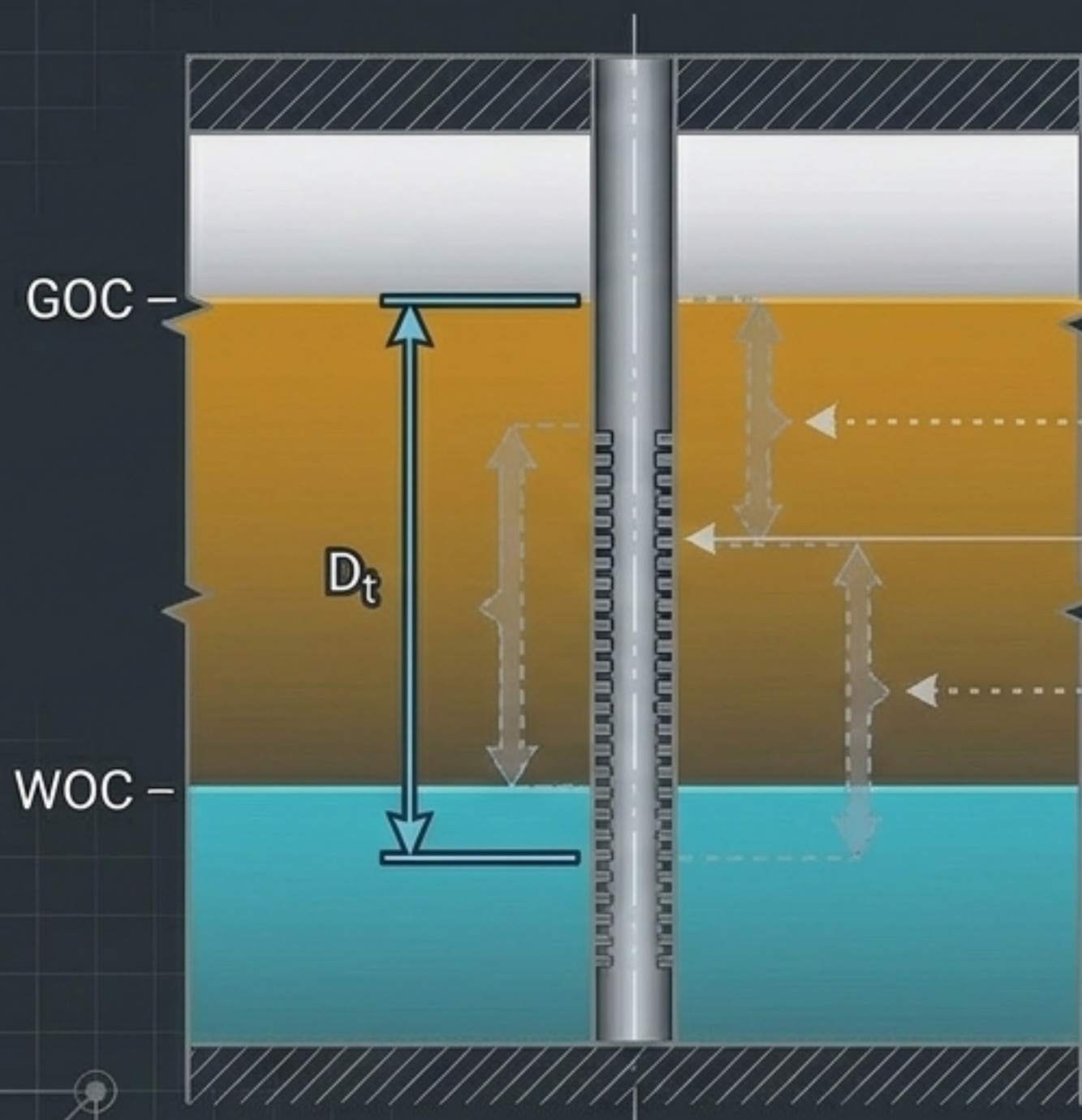


The Squeeze: When a thin oil zone is overlaid by a gas cap and underlaid by bottom water, the completion interval (h_p) must be perfectly calibrated to permit maximum oil flow without triggering a dual breakthrough.

Pirson's Combined Correlation (1977):

$$Q_{oc} = 0.246 * 10^{-4} \left[\frac{k_o}{u_o B_o \ln(r_e/r_w)} \right] (h^2 - h_p^2) \cdot \left[(p_w - p_o) \left(\frac{p_o - p_g}{p_w - p_g} \right)^2 + (p_o - p_g) \left(1 - \frac{p_o - p_g}{p_w - p_g} \right)^2 \right]$$

Strategic Placement: Finding the Optimum Interval



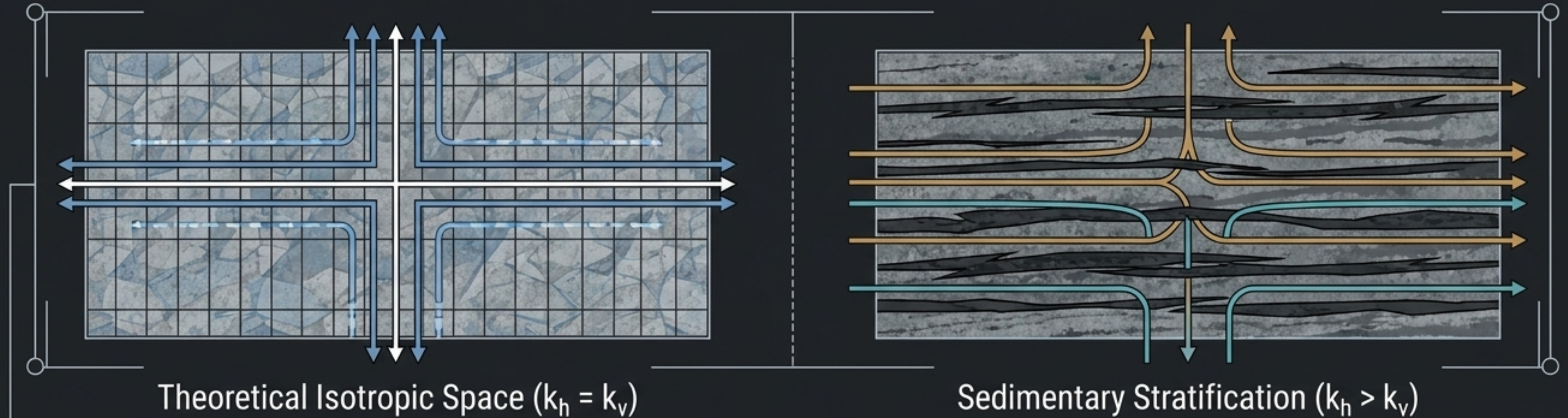
Operational Goal: Calculating the exact depth (D_t) that perfectly balances the competing gravitational pulls of the top gas cap and bottom water zone.

Pirson's Optimum Placement Equation (9-4):

$$D_t = (h - h_p) \left[1 - \frac{p_o - p_g}{p_w - p_g} \right]$$

Tactical Insight: Optimum placement relies entirely on the interplay of fluid densities (ρ) and the total unperforated oil column ($h - h_p$). Production engineers can actively shift D_t based on these specific fluid weight ratios to delay dual breakthrough.

Reality Check: The Flaw in the Homogeneous Assumption



The Flaw (Slider's Critique): Equations 9-1 through 9-4 fundamentally assume permeability is identical in all directions.

The Reality: Sedimentary formations are laid down in thin horizontal sheets. Microscopic, impermeable shale barriers drastically choke vertical flow while having little effect on horizontal flow.

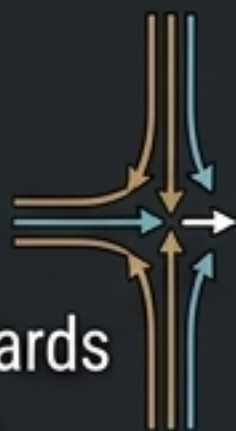
The Result: Horizontal permeability (k_h) is almost always significantly greater than vertical permeability (k_v).

Evolution (Chierici-Ciucci): Advanced potentiometric models evolved to use dimensionless graphs specifically to account for this actual k_v/k_h ratio constraint.

Managing the Subsurface Threat: A Unified Framework

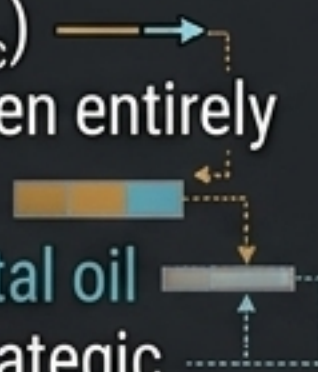
The Mechanism

Coning is the dynamic consequence of viscous pressure gradients overpowering static gravitational separation, driving gas downwards and water upwards into the pressure sink.



The Baseline Math

The Meyer-Garder and Pirson correlations provide the foundational critical rate (Q_{oc}) thresholds, driven entirely by fluid density differentials, total oil column, and strategic perforation placement (D_t).



The Geological Reality

True reservoir optimization requires moving beyond isotropic assumptions. Engineers must account for sedimentary stratification ($k_h > k_v$) to accurately predict and delay breakthrough.

